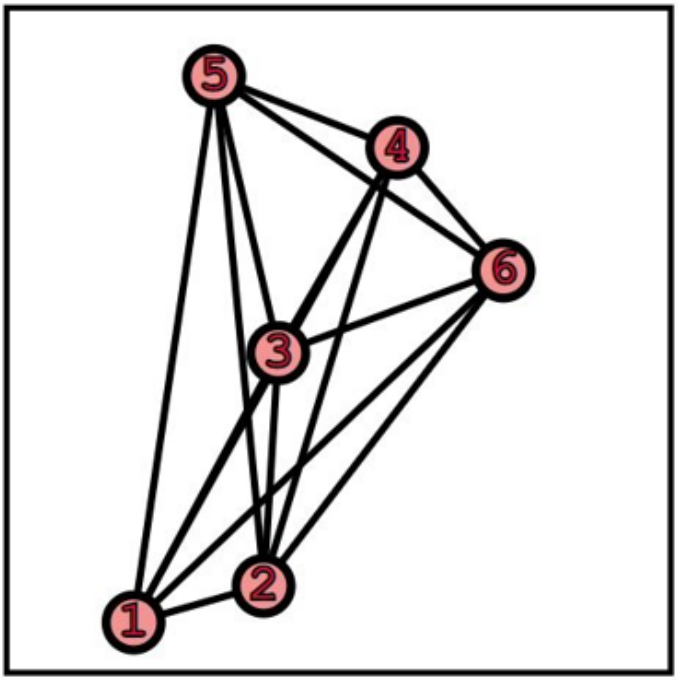


Figure 16.3 A random embedding for the K5 complete graph. Notice how particularly bad the choice of positions turned out to be.

Figure 16.2 shows a few examples of valid and invalid choices of vertex centers, given our constraints; in particular we will have to check the conditions at points 4 and 5, once we have chosen all the points. We'll defer the discussion about checking issues #4 and #5; for now, we can assume we have helper methods for these checks, and if we find out any of these constraints is violated, then we have two strategies to make it right:

- Correct the vertices position, for example slightly perturbating, at random, those points that coincide or intersect an edge;
- Discard the solution violating a constraint and start over.

Notice that, if we draw vertices as circles (like in figure 16.2) and not just points, we will have to make the requirement on vertices stronger, requiring that the two circles don't intersect.